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4.5/10

COURSE: CHE 312: Transport Phenomena II

Spring Semester 2020

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Quiz #2

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Closed Notes and Book

Duration: 25 minutes

A nuclear reactor fuel element consists of a solid cylindrical pin of radius r_f and thermal conductivity k_f. The fuel pin is in good contact with a cladding material of outer radius r_c and thermal conductivity k_c Consider steady-state conditions for which uniform heat generation occurs within the fuel at a volumetric rate $\dot{q}$ and the outer surface of the cladding is exposed to a coolant that is characterized by a temperature T_∞ and a convection coefficient h.

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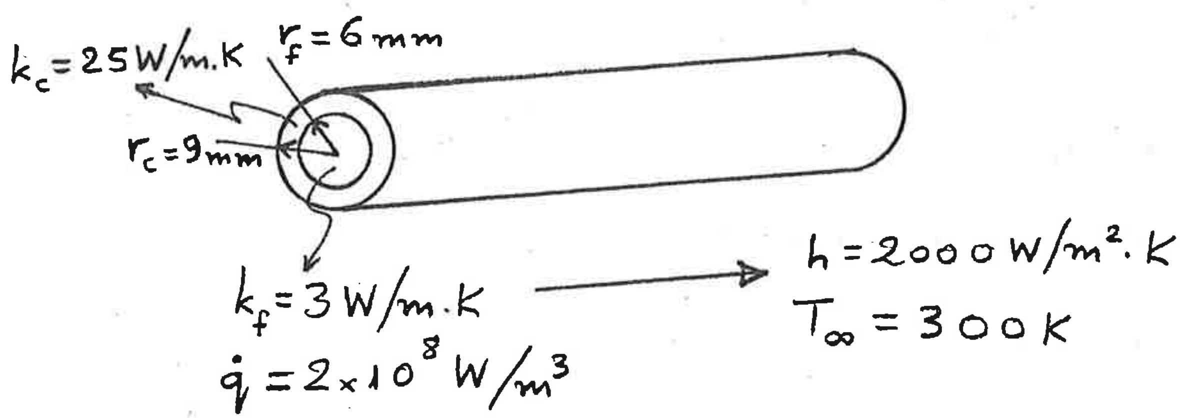
a) Obtain equations for the temperature distributions $T_f(r)$ and $T_c(r)$ in the fuel and cladding, respectively. Express your results exclusively in terms of the foregoing variables.

b) Consider a uranium oxide fuel pin for which k_f = 3 W/m·K and r_f = 6 mm and cladding for which k_c = 25 W/m·K and r_c = 9 mm. If $\dot{q} = 2 \times 10^8$ W/m^3, h = 2000 W/m^2·K, and T_∞ = 300 K, what is the maximum temperature in the fuel element?



Steady State:

$q_{conv} = q_c = q_f$

a.)

$\int \partial \left(kr \frac{\partial T}{\partial r} \right) = \int -\dot{q} r \, dr \rightarrow kr \frac{\partial T}{\partial r} = -\frac{1}{2} \dot{q} r^2 + C_1$

$\int k \, \partial T = \int -\frac{1}{2} \dot{q} r + \frac{C_1}{r} \, dr$

$kT = -\frac{1}{4} \dot{q} r^2 + C_1 \ln|r| + C_2 \rightarrow T(r) = -\frac{1}{4k} \dot{q} r^2 + \frac{C_1}{k} \ln|r| + C_2$

B.C @ $r = r_f$ $T = T_f$ @ $r = r_c$, $T = T(c)$ ✗

BCs
(i) $r = 0 \Rightarrow \frac{dT_f}{dr} = 0$
(ii) $r = r_f \Rightarrow T_f(r_f) = T_c(r_f)$
& so on...

$T_f = -\frac{1}{4k_f} \dot{q} r_f^2 + \frac{C_1}{k_f} \ln|r_f| + C_2$

$T_c = -\frac{1}{4k_c} \dot{q} r_c^2 + \frac{C_1}{k_c} \ln|r_c| + C_2$ ✗

Subtract

Not needed

$T_f - T_c = -\frac{1}{4} \dot{q} \left(r_f^2 + r_c^2 \right) + C_1 \left[\left(\frac{1}{k_f} - \frac{1}{k_c} \right) \ln \left| \frac{r_f}{r_c} \right| \right]$

$C_1 = \dfrac{T_f - T_c + \frac{1}{4} \dot{q} \left(r_f^2 + r_c^2 \right)}{\left(\frac{1}{k_f} - \frac{1}{k_c} \right) \ln|r_f / r_c|}$

$$\boxed{\begin{aligned} T_f(r) &= -\frac{1}{4k_f} \dot{q} r_f^2 + \frac{T_f - T_c + \frac{1}{4} \dot{q} \left(r_f^2 + r_c^2 \right) \ln|r_f|}{\left[\left(\frac{1}{k_f} - \frac{1}{k_c} \right) \ln|r_f / r_c| \right] \cdot k_f} \cancel{+ C_2} \\ T_c(r) &= -\frac{1}{4k_c} \dot{q} r_c^2 + \frac{T_f - T_c + \frac{1}{4} \dot{q} \left(r_f^2 + r_c^2 \right) \ln|r_c|}{\left[\left(\frac{1}{k_f} - \frac{1}{k_c} \right) \ln|r_f / r_c| \right] \cdot k_c} \end{aligned}}$$

Make $C_2 = 0$ ✗
why?

b.) Steady State $q_{conv} = q_c = q_f$

$h(T_\infty - T_c) = -k \frac{T_\infty - T_c}{r_c} \Rightarrow 2000 \, W/m^2 K \cdot (300K - T_c) = \frac{(25 \, W/m \, K)(300K - T_c)}{.009 \, mm}$

$6 \times 10^5 - 2000 T_c = 8.33 \times 10^5 - 2.77 \times 10^3 T_c$ ✗

$777 T_c = 2.33 \times 10^5$

$\boxed{T_c = 299.87^\circ C}$ $T_f =$ ✗